Sovereign Debt Signaling with Central Bank Balance Sheet Policies

This Version: April 27, 2026

The goal of this note is to show another implication of risk-averse banks: inflation determination.

Model with Bonds, Reserves, and Balance Sheet Policies

A potential way to extend the baseline signaling framework in your paper to an environment with a fiscal authority, a central bank, and segmented financial markets.

The goal of such an environment would be to study how sovereign borrowing signals private information when government debt interacts with central-bank balance-sheet policies.

In this new environment, time is discrete and the economy lasts for three periods, $t = 0, 1, 2$. Agents have access to assets, which they are all nominal. The fiscal authority issues government bonds are nominal claims issued in period 0 that mature in period 2, while reserves are one-period nominal liabilities issued by the central bank. The aggregate price level in period t is denoted by P_t.

There are four types of agents:

- the fiscal authority,
- the central bank,
- reserve-access agents ("banks"),
- non-reserve-access agents ("non-banks").

The measure of banks is $\mu \in (0, 1)$ and the measure of non-banks is $1 - \mu$.

Fiscal Authority

The fiscal authority issues nominal government bonds in period 0. Let $B \in \mathcal{B}$ denote the face value of nominal bonds issued in period 0. These bonds promise repayment of B units of currency in period 2.

Let $Q_0^B(B)$ denote the nominal price, in units of currency at $t = 0$, of one unit of face value of government bonds maturing in period 2. The fiscal authority's nominal budget constraint in period 0 is given by

$$P_0 C_0^F = P_0 X + Q_0^B(B)B + T_0^{CB},$$

where C_0^F denotes period-0 real fiscal absorption, X is the period-0 real fiscal endowment, and T_0^{CB} is the nominal transfer from the central bank.

In period 2, the fiscal authority receives the nominal value of fiscal resources

$$P_2(Y + \phi),$$

where Y is a stochastic real endowment and ϕ captures future real fundamentals.

If the fiscal authority repays its nominal debt obligations, its period-2 nominal budget constraint is given by

$$P_2 C_2^{F,R} = P_2(Y + \phi) + T_2^{CB,R} - B.$$

If it defaults, it obtains the fallback payoff

$$P_2 C_2^{F,D} = P_2 \kappa + T_2^{CB,D},$$

where κ is the real payoff under default. Bondholders receive zero in default. Thus one unit of government bond face value pays one unit of currency in repayment and zero in default.

The government compares the repayment payoff, which uses the remittance $T_2^{CB,R}$ induced by repayment, to the default payoff, which uses the remittance $T_2^{CB,D}$ induced by default. Default occurs whenever

$$P_2(Y + \phi) + T_2^{CB,R} - B < P_2 \kappa + T_2^{CB,D}.$$

Thus, the default probability for type i is

$$D\left(\kappa_i + \frac{B - (T_2^{CB,R} - T_2^{CB,D})}{P_2} - \phi_i\right).$$

The fiscal authority derives utility from real fiscal resources. For type $i \in \{S, R\}$,

$$u_i(B) = \ln\left(\frac{P_0 X + Q_0^B(B)B + T_0^{CB}}{P_0}\right) + D\left(\kappa_i + \frac{B - (T_2^{CB,R} - T_2^{CB,D})}{P_2} - \phi_i\right) \ln\left(\kappa_i + \frac{T_2^{CB,D}}{P_2}\right)$$

$$+ \left[1 - D\left(\kappa_i + \frac{B - (T_2^{CB,R} - T_2^{CB,D})}{P_2} - \phi_i\right)\right] \mathbb{E}\left[\ln\left(Y + \phi_i + \frac{T_2^{CB,R} - B}{P_2}\right) \middle| P_2(Y + \phi_i) + T_2^{CB,R} - B >\right.$$

As in the baseline model, the fiscal authority receives idiosyncratic choice shocks $\varepsilon(B)$ drawn from a type-I extreme value distribution. The borrowing choice satisfies

$$B_i = \arg\max_{B \in \mathcal{B}} \{u_i(B) + \varepsilon(B)\}.$$

Central Bank

The central bank issues nominal reserves, pays nominal interest on reserves, conducts open market operations, and transfers its net proceeds to the fiscal authority. The policy instruments are the reserve supplies $\{M_0, M_1\}$, the bond purchases $\{B_0^{CB}, B_1^{CB}\}$, and the reserve rates $\{i_0^R, i_1^R\}$. The transfers $\{T_0^{CB}, T_2^{CB,R}, T_2^{CB,D}\}$ are residual objects implied by the central bank's budget constraints, while the period-1 transfer is normalized to zero.

Let M_0 denote nominal reserves issued in period 0 and M_1 nominal reserves issued in period 1. Reserves are one-period nominal liabilities held only by banks. Let i_t^R denote the nominal interest rate paid on reserves issued in period t. Let B_0^{CB} denote the nominal face value of government bonds purchased by the central bank in period 0, and let B_1^{CB} denote additional nominal bond purchases in period 1.

Period 0 budget constraint: In period 0, the central bank finances its purchases of nominal government bonds and its transfer to the fiscal authority by issuing reserves. Its nominal flow budget constraint is then given by

$$Q_0^B(B)B_0^{CB} + T_0^{CB} = M_0.$$

Hence the period-0 transfer is the residual

$$T_0^{CB} = M_0 - Q_0^B(B)B_0^{CB}.$$

Period 1 budget constraint: In period 1, reserves issued in period 0 mature and must be repaid with nominal interest. The central bank may issue new nominal reserves M_1 and may purchase additional nominal government bonds B_1^{CB}. We normalize the period-1 transfer to the fiscal authority to zero, so $T_1^{CB} = 0$.

Let Q_1^B denote the nominal period-1 price of government bonds maturing in period 2. The period-1 nominal flow budget constraint is

$$Q_1^B B_1^{CB} + (1 + i_0^R)M_0 = M_1.$$

Period 2 profits and transfers: By period 2, the central bank holds the following nominal bonds

$$B^{CB} = B_0^{CB} + B_1^{CB}.$$

If the government repays, the central bank receives the nominal payoff that is B^{CB}.

The central bank must also repay period-1 reserves with nominal interest:

$$(1 + i_1^R)M_1.$$

Hence, if the government repays, period-2 nominal profits for the central bank are

$$\Pi_2^{CB} = B^{CB} - (1 + i_1^R)M_1.$$

These profits are remitted to the fiscal authority so that

$$T_2^{CB,R} = \Pi_2^{CB}.$$

If the government defaults, the central bank's bond portfolio pays zero, so the period-2 remittance is

$$T_2^{CB,D} = -(1 + i_1^R)M_1.$$

Under zero recovery, the remittance difference across the repayment and default contingencies is therefore

$$T_2^{CB,R} - T_2^{CB,D} = B^{CB}.$$

Let $\mathbf{1}_2^R(B)$ denote the indicator that the government repays in period 2 after issuing B, so that the realized nominal payoff of one unit of bond face value is $\mathbf{1}_2^R(B)$.

Private Agents

Private agents differ only in their access to central bank reserves. Both types of agents form beliefs about the fiscal authority's type $i \in \{S, R\}$ and price government bonds accordingly.

The government type i captures private information relevant for default risk. Two interpretations are considered:

- **Hidden default payoff**: the type differs in the nominal fallback payoff κ_i.
- **Hidden fundamentals**: the type differs in future real fiscal resources ϕ_i.

Neither banks nor non-banks observe the realization of i. Instead, they observe government borrowing B and update beliefs using Bayes' rule.

Reserve-access Agents (Banks)

Banks have access to central bank reserve accounts and can therefore hold both nominal reserves and government bonds. Let B_t^A denote the nominal face value of government bonds held by banks at time t, and let M_t denote their nominal reserve holdings. Banks are competitive, risk-neutral intermediaries. Since they can substitute at the margin between reserves and government bonds, equilibrium requires indifference between the two assets whenever both are held.

Their period-0 nominal budget constraint is

$$P_0 C_0^A + Q_0^B(B) B_0^A + M_0 = P_0 W_0^A.$$

Similarly, for period 1 we have that

$$P_1 C_1^A + Q_1^B B_1^A + M_1 = P_1 W_1^A + (1 + i_0^R) M_0 + Q_1^B B_0^A.$$

In period 2, the bank's nominal budget constraint is

$$P_2 C_2^A = (1 + i_1^R) M_1 + \mathbf{1}_2^R(B) B_1^A.$$

Let $\mathbb{E}_t[\cdot]$ denote expectations conditional on information at time t, including posterior beliefs about the fiscal authority's type after observing borrowing B. Bank indifference implies the asset-pricing conditions

$$Q_1^B = \frac{\mathbb{E}_1[\mathbf{1}_2^R(B)]}{1 + i_1^R},$$

and

$$Q_0^B(B) = \frac{\mathbb{E}_0[Q_1^B \mid B]}{1 + i_0^R}.$$

These conditions make reserve rates the marginal opportunity cost of bond holdings for banks.

Non-Reserve-Access Agents (Non-Banks)

Non-banks do not have access to reserves and therefore can only hold nominal government bonds. Let B_t^N denote their nominal bond holdings. They choose consumption and bond holdings to smooth consumption over time. Specifically, they solve

$$\max_{\{C_t^N, B_t^N\}} \; u(C_0^N) + \beta u(C_1^N) + \beta^2 \mathbb{E}_0[u(C_2^N)]$$

subject to their budget constraints. Their period-0 nominal budget constraint is

$$P_0 C_0^N + Q_0^B(B) B_0^N = P_0 W_0^N$$

while for period 1 is given by

$$P_1 C_1^N + Q_1^B B_1^N = P_1 W_1^N + Q_1^B B_0^N.$$

In period 2, their nominal budget constraint is

$$P_2 C_2^N = \mathbf{1}_2^R(B) B_1^N.$$

The associated Euler equations are

$$\frac{Q_0^B(B)}{P_0} u'(C_0^N) = \beta \mathbb{E}_0 \left[\frac{Q_1^B}{P_1} u'(C_1^N) \,\middle|\, B \right],$$

and

$$\frac{Q_1^B}{P_1} u'(C_1^N) = \beta \mathbb{E}_1 \left[\frac{\mathbf{1}_2^R(B)}{P_2} u'(C_2^N) \right].$$

Like banks, non-banks do not observe the government type i. They form posterior beliefs $\rho(B)$ based on observed borrowing and evaluate expected bond payoffs using those beliefs.

Role of Financial Segmentation: The distinction between banks and non-banks affects bond pricing because only banks can substitute between reserves and government bonds. Consequently, changes in reserve supply and interest on reserves influence banks' portfolio choices and therefore affect equilibrium bond prices.

However, both banks and non-banks share the same belief about the government's type i and evaluate default risk using the same posterior probability $\rho(B)$.

Market Clearing

Bond markets clear in each period. In period 0, we have that

$$B = B_0^{CB} + \mu\, B_0^A + (1-\mu)\, B_0^N.$$

In period 1, the outstanding stock of bonds is allocated across the central bank, banks, and non-banks according to

$$B = B_0^{CB} + B_1^{CB} + \mu\, B_1^A + (1-\mu)\, B_1^N.$$

Reserves are held only by banks. Let m_t denote per-bank reserve holdings. Since the central bank issues aggregate reserves M_t, reserve-market clearing implies

$$M_0 = \mu\, m_0, \qquad M_1 = \mu\, m_1.$$

Goods-market clearing holds period by period in real terms.

Government Types and Private Information

As in the baseline model, there are two fiscal-authority types $i \in \{S, R\}$. Two alternative sources of private information are considered.

Hidden default payoff: The government type differs in the real fallback payoff:

$$\kappa_R > \kappa_S.$$

A higher κ corresponds to a less severe effective punishment from default.

Hidden future fundamentals: The government type differs in future real fundamentals:

$$\phi_S > \phi_R.$$

This captures private information about future fiscal capacity, growth, or the productivity of debt-financed expenditures.

Default Probability

Let Y be distributed with cumulative distribution function $D(\cdot)$. Then the default probability for type i is given by

$$\Pr(\text{default} \mid i, B) = D\left(\kappa_i + \frac{B - B^{CB}}{P_2} - \phi_i\right).$$

Higher nominal bond issuance increases default risk, while larger central-bank bond holdings reduce default risk by raising the remittance loss associated with default.

Beliefs and Bond Pricing

Creditors do not observe the government type, but they observe nominal borrowing B. Let the prior probability of the safe type be π. Posterior beliefs satisfy

$$\rho(B) = \frac{\pi \Pr(B|S)}{\pi \Pr(B|S) + (1-\pi)\Pr(B|R)}.$$

Under the zero-recovery assumption, the expected nominal payoff of one unit of government bonds in period 2 is just the repayment probability under posterior beliefs:

$$\mathbb{E}_1[\mathbf{1}_2^R(B)] = 1 - \bar{D}_1(B).$$

Because banks are indifferent at the margin between reserves and government bonds, bond prices satisfy

$$Q_1^B = \frac{1 - \bar{D}_1(B)}{1 + i_1^R},$$

and

$$Q_0^B(B) = \frac{\mathbb{E}_0[Q_1^B \mid B]}{1 + i_0^R}.$$

Thus reserve rates enter bond pricing through bank arbitrage, while posterior beliefs determine bond prices through repayment probabilities.

Equilibrium

Given central-bank policy instruments $\{M_0, M_1, B_0^{CB}, B_1^{CB}, i_0^R, i_1^R\}$, an equilibrium consists of

- fiscal-authority borrowing choices B_i,
- residual central-bank transfers $\{T_0^{CB}, T_2^{CB,R}, T_2^{CB,D}\}$ implied by the central-bank budget constraints,
- bank portfolio choices $\{B_0^A, B_1^A\}$,
- non-bank allocations $\{C_0^N, C_1^N, C_2^N, B_0^N, B_1^N\}$,
- nominal bond prices $\{Q_0^B, Q_1^B\}$,
- posterior beliefs $\rho(B)$,

such that

- the fiscal authority optimizes,
- the central-bank budget constraints hold and determine residual transfers,
- banks satisfy the reserve-bond indifference conditions,
- non-banks solve their consumption-smoothing problem,
- bond and reserve markets clear, and
- beliefs satisfy Bayes' rule.

Equilibrium Characterization

Effect of Reserve Rates

The equilibrium can be characterized in a simple way. Banks are the marginal arbitrageurs in the bond market because they are the only agents that can substitute between reserves and government bonds. As a result, reserve rates pin down bond prices through the bank indifference conditions. Non-banks instead take bond prices as given and choose their bond holdings to smooth consumption over time.

Given these prices, non-bank bond demand is determined by their Euler equations, while the central bank's bond holdings are fixed by policy. Market clearing then implies that banks absorb the residual supply of government bonds. Thus, in the current specification, banks determine bond prices at the margin, whereas non-banks affect the equilibrium allocation of bonds across sectors without independently determining the price.

Effect of Reserve Supply

Reserve supply plays a more limited role in the current specification. Period-0 reserves affect the government's borrowing decision because M_0 changes the period-0 transfer T_0^{CB} and therefore changes the government's initial resources. By contrast, once the period-1 transfer is normalized to zero, the period-1 central-bank budget constraint implies M_1 from the choices of M_0, B_1^{CB}, and the prevailing bond price.

As a result, reserve supply does not directly pin down bond prices, which are instead determined by reserve rates through bank arbitrage. Moreover, M_1 does not directly affect default probabilities in the current zero-recovery specification because it enters the repayment-state and default-state remittances symmetrically and therefore cancels out of the government's repayment-default comparison. Thus, in the present model, M_0 matters through the government's initial financing, whereas M_1 is mainly a balance-sheet object implied by the central bank's period-1 budget constraint.

Effect of Central-Bank Bond Purchases

Central-bank bond purchases matter primarily through the default-risk channel. Both period-0 purchases B_0^{CB} and period-1 purchases B_1^{CB} increase total central-bank bond holdings carried into period 2. Under the current zero-recovery specification, this raises the remittance loss associated with sovereign default and therefore lowers default probabilities.

By contrast, the portfolio-absorption effect of central-bank purchases has no independent equilibrium consequence in the current model. Since banks are residual holders and bond prices are pinned down by reserve rates through bank arbitrage, a change in the quantity of bonds held by the central bank does not by itself create a separate price effect through market clearing. Any effect of period-1 purchases on the period-1 bond price arises only because lower default risk raises expected repayment. Likewise, anticipated future purchases can affect the period-0 bond price only through their effect on expected future repayment and expected future bond prices.

Market Segmentation in Previous Framework

Even though the model has segmentation whereby banks can hold reserves and bonds and non-banks can hold bonds only; there is still no *"pricing wedge"* because: (i) banks are the marginal arbitrageurs, and (ii) non-banks do not affect bond pricing directly. These features are a direct consequence of banks being <u>risk neutral</u>.

To introduce a meaningful *"segmentation wedge in asset prices"*, the environment needs one of the following features: (i) risk averse banks; (ii) balance sheet constraints; or (iii) preferred habitat demand; among other potential features.

Risk Averse Banks

Let us now consider banks to be risk averse as it is the case with non-banks. This is a way to capture that the bank's clients are risk averse, without having to model them. One could think the bank as an ex-ante coalition of agents as in Diamond and Dybvig (1983).

When banks are risk averse, reserves and government bonds are no longer perfect substitutes at the margin. Although banks can hold both assets, bonds expose them to sovereign default risk while reserves do not. As a result, banks require a risk premium to absorb government bonds. Because non-banks cannot hold reserves, this premium is not arbitraged away and appears as a wedge between the reserve rate and the bond yield. With risk averse banks, then a *"wedge due to financial segmentation"* emerges and central bank purchases can have a portfolio balance effect on the economy.

Bond Pricing with Risk-Averse Banks

I now *"sketch"* how this might work. Let Λ_{t+1}^{A} denote the nominal stochastic discount factor of banks between periods t and $t+1$. Bank portfolio choices imply the following asset pricing conditions.

At time 1, from the bank's perspective, the price of a nominal government bond maturing in period 2 satisfies the following

$$Q_1^B = E_1\left[\Lambda_2^A \, \mathbf{1}_2^R(B)\right],$$

where $\mathbf{1}_2^R(B)$ denotes the realized nominal payoff of one unit of government bond face value in period 2.

Reserves are one period nominal liabilities of the central bank that pay the gross nominal return $(1+i_1^R)$ with certainty. Hence from the bank's perspective, their pricing condition is given by

$$1 = E_1\left[\Lambda_2^A(1+i_1^R)\right].$$

Combining the two conditions yields

$$Q_1^B = \frac{E_1\left[\Lambda_2^A \mathbf{1}_2^R(B)\right]}{E_1[\Lambda_2^A(1+i_1^R)]}.$$

Since bond payoffs are low precisely in states where marginal utility is high (sovereign default states), banks require compensation for bearing this risk. A convenient reduced-form representation is therefore

$$Q_1^B = \frac{1-\bar{D}_1(B)}{1+i_1^R+\tau_1},$$

where $\bar{D}_1(B)$ denotes the expected default probability under posterior beliefs and $\tau_1 > 0$ represents a risk premium reflecting banks' risk aversion and balance-sheet exposure to sovereign bonds.

Similarly, the period-0 bond price satisfies

$$Q_0^B(B) = \frac{E_0\left[Q_1^B \mid B\right]}{1+i_0^R+\tau_0}.$$

The terms τ_0 and τ_1 capture wedges between reserve rates and sovereign yields that arise because only banks can arbitrage between reserves and bonds and because banks require compensation to absorb risky sovereign debt.

With risk averse banks, bond prices now depend on three factors: (i) expected sovereign repayment probabilities, (ii) reserve rates set by the central bank, and (iii) the risk premium required by banks to absorb sovereign debt.

Effect of Central-Bank Bond Purchases

With risk averse banks, central bank bond purchases influence bond prices through two channels. First, as in the baseline specification, higher central bank bond holdings increase the remittance loss associated with sovereign default and therefore reduce default probabilities. Second, purchases reduce the quantity of risky sovereign bonds that must be held by risk averse banks. By lowering the exposure of bank balance sheets to sovereign risk, central bank purchases reduce the risk premium and raise bond prices. This second mechanism corresponds to the portfolio-balance channel of monetary policy.

Risk Aversion and Nominal Determination

Beyond the pricing wedge and portfolio balance channel, risk aversion in the banking sector has a further consequence: it partially resolves the nominal indeterminacy of the risk-neutral specification. With risk-neutral banks, the price level path $\{P_0, P_1, P_2\}$ is entirely free. With risk-averse banks, inflation rates are pinned down, leaving only the initial price level P_0 as a free normalization.

Source of indeterminacy with risk-neutral banks. With constant marginal utility, banks are indifferent about the timing of consumption. Their sole equilibrium role is to arbitrage between reserves and bonds, pinning down Q_t^B independently of $\{P_t\}$. Bank consumption in each period is a free residual: for any price level path, C_t^A adjusts to absorb whatever remains after fiscal and non-bank consumption, satisfying goods-market clearing trivially. All three price levels are therefore unconstrained.

Goods-market clearing conditions. With real endowments W_t^A, W_t^N and real default payoff κ, the aggregate resource constraints can be derived period by period by summing all agents' budget constraints and imposing bond and reserve market clearing.

Period 0. Summing the fiscal budget constraint ($P_0 C_0^F = P_0 X + Q_0^B B + T_0^{CB}$ with $T_0^{CB} = M_0 - Q_0^B B_0^{CB}$), μ times the per-bank budget constraint, and $(1-\mu)$ times the per-non-bank budget constraint, and using bond market clearing $B = B_0^{CB} + \mu\, B_0^A + (1-\mu)\, B_0^N$ and reserve market clearing $M_0 = \mu\, m_0$, all nominal asset terms cancel, yielding

$$C_0^F + \mu\, C_0^A + (1-\mu)\, C_0^N = X + \mu\, W_0^A + (1-\mu)\, W_0^N.$$

Period 1*.* The fiscal authority has no endowment or consumption in period 1. Summing μ times the per-bank and $(1-\mu)$ times the per-non-bank period-1 budget constraints and using bond market clearing $\mu\, B_1^A + (1-\mu)\, B_1^N = B - B_0^{CB} - B_1^{CB}$, reserve market clearing, and the central bank period-1 budget constraint $M_1 = Q_1^B B_1^{CB} + (1+i_0^R) M_0$, all nominal asset terms cancel, yielding

$$\mu\, C_1^A + (1-\mu)\, C_1^N = \mu\, W_1^A + (1-\mu)\, W_1^N.$$

Period 2 *(repayment).* Summing the fiscal, μ times the per-bank, and $(1-\mu)$ times the per-non-bank period-2 budget constraints under repayment and using bond market clearing $\mu\, B_1^A + (1-\mu)\, B_1^N = B - B^{CB}$ and the central bank transfer $T_2^{CB,R} = B^{CB} - (1+i_1^R) M_1$, all nominal terms cancel, yielding

$$C_2^{F,R} + \mu\, C_2^A + (1-\mu)\, C_2^N = Y + \phi.$$

Under default, the same derivation gives $C_2^{F,D} + \mu\, C_2^A + (1-\mu)\, C_2^N = \kappa$, which is also purely real since κ is a real quantity.

Because W_t^A, W_t^N, X, Y, ϕ, and κ are all real, the right-hand side of each goods-market clearing condition is independent of the price level path $\{P_t\}$.

Inflation is determined but the price level is not. With risk-averse banks maximizing $u(C_0^A) + \beta\, u(C_1^A) + \beta^2\, \mathbb{E}_0[u(C_2^A)]$ subject to their budget constraints, the first-order condition for reserves yields

$$\frac{u'(C_t^A)}{P_t} = \beta(1+i_t^R)\, \mathbb{E}_t\left[\frac{u'(C_{t+1}^A)}{P_{t+1}}\right],$$

which, since $u'' < 0$, pins down bank consumption across adjacent periods as a function of the inflation rate P_{t+1}/P_t and the reserve rate i_t^R. Analogous Euler equations pin down non-bank consumption, while C_t^F is determined by the fiscal authority's budget constraint. Since each C_t^j on the left-hand side of the goods-market clearing conditions depends on inflation rates, and the right-hand side is a fixed real quantity, the goods-market clearing conditions determine the inflation rates P_1/P_0 and P_2/P_1 that equate aggregate consumption to the available endowment in each period.

However, the initial price level P_0 is not determined. Scaling all nominal variables (P_0, P_1, P_2 and all nominal asset quantities) by a common factor $\lambda > 0$ leaves all real allocations unchanged: inflation rates, real consumption, real bond prices Q_t^B/P_t, and real reserve holdings M_t/P_t are all invariant. Thus P_0 is a free normalization.

Pass-through of reserve rate to inflation: Because inflation is now endogenous, reserve rates affect the real value of government debt. The Euler equations, budget constraints, and goods-market clearing conditions jointly determine the consumption sequence, real interest rates, and inflation rates. The bank's reserve Euler equation implies the Fisher relation

$$\frac{P_{t+1}}{P_t} = \frac{1 + i_t^R}{1 + r_t}, \qquad 1 + r_t \equiv \frac{u'(C_t^A)}{\beta\, \mathbb{E}_t[u'(C_{t+1}^A)]}.$$

When the central bank raises i_t^R, the real rate r_t may also adjust through general equilibrium effects: banks shift consumption toward the future, non-banks absorb the opposite shift, and bond prices adjust. However, aggregate real consumption in each period is tied to the fixed real endowment by goods-market clearing. This limits how much r_t can move and ensures that most of the increase in i_t^R passes through to higher inflation P_{t+1}/P_t rather than higher r_t.

The pass-through rate from i_t^R to inflation is monotonically increasing in three quantities:

- *Bank share of aggregate consumption* (μ). When banks are the entire economy ($\mu = 1$), goods-market clearing fixes C_t^A at the endowment, the real rate is entirely independent of i_t^R, and the pass-through is one-for-one. When banks are negligible ($\mu \to 0$), they freely adjust their consumption profile without disturbing goods-market clearing, r_t absorbs the full change in i_t^R, and the pass-through vanishes.
- *Bank risk aversion* (γ^A). More risk-averse banks are less willing to shift consumption intertemporally, so r_t moves less for a given change in i_t^R. In the limit $\gamma^A \to 0$ (risk neutrality), banks freely adjust consumption and the nominal indeterminacy of the risk-neutral specification is recovered.

- *Non-bank risk aversion* (γ^N). More risk-averse non-banks resist absorbing the consumption shift that banks attempt, forcing more of the adjustment through inflation rather than through redistribution of consumption across agent types.

Reserve issuance and inflation. Under the risk-averse-bank extension, the right way to think about reserve issuance is that it can affect inflation only if it changes the bank-implied real rate. There is no mechanical "more reserves, more inflation" result. The key relation remains

$$\frac{P_{t+1}}{P_t} = \frac{1+i_t^R}{1+r_t}, \qquad 1+r_t \equiv \frac{u'(C_t^A)}{\beta\, \mathbb{E}_t[u'(C_{t+1}^A)]}.$$

Therefore, M_t affects inflation only if it moves r_t, meaning only if it changes banks' consumption growth or risk exposure.

A general-equilibrium wealth or portfolio channel is possible through three linked mechanisms:

- *Intertemporal wealth transfer.* Higher M_0 raises the period-0 transfer because

  $$T_0^{CB} = M_0 - Q_0^B(B)B_0^{CB}.$$

  But the central bank must later service those reserves, since

  $$M_1 = Q_1^B B_1^{CB} + (1+i_0^R)M_0,$$

  and this shows up as lower future remittances. Hence reserve issuance shifts public resources across time: more spending, or less Treasury borrowing, today, but more reserve servicing tomorrow. If that shift makes banks consume relatively less today and more tomorrow, then r_0 rises and P_1/P_0 falls for a given reserve rate. If it does the opposite, inflation rises.
- *Portfolio-risk reallocation.* With risk-averse banks, reserves are safe and sovereign bonds are risky. If reserve issuance is tied to central-bank bond purchases or to lower Treasury borrowing, banks end up holding a safer portfolio. Since the draft already summarizes this force through a lower risk premium τ_t, a safer balance sheet can reduce precautionary saving, shift bank consumption toward the present, lower r_t, and raise inflation for a given reserve

rate. However, the sign is not automatic and depends on how the full equilibrium allocation adjusts.

- *Debt-default feedback.* If more reserve issuance at $t = 0$ lets the fiscal authority issue less debt B, then future sovereign risk changes. That alters bond prices, bank wealth in repayment and default states, and expected marginal utilities. Since inflation is pinned down through those marginal utilities together with goods-market clearing, reserve issuance can feed back into inflation indirectly through the sovereign-risk block as well.

Therefore, there is a clean reserve-rate channel to inflation, but only a potential reserve-quantity channel, operating through wealth redistribution and portfolio composition. The sign of that quantity channel is not pinned down in the current text. The reason is that the aggregate resource constraints cancel nominal asset quantities out, so reserve issuance does not determine inflation the way a quantity-theory model would. Instead, reserve issuance is best interpreted as public-liability composition policy rather than pure money creation: it matters only if changing the mix of safe reserves and risky sovereign bonds changes the bank's desired intertemporal consumption path.

Inflationary finance channel: Higher inflation raises P_2 and reduces the real value of nominal debt $(B - B^{CB})/P_2$ in the default probability

$$D\left(\kappa_i + \frac{B - B^{CB}}{P_2} - \phi_i\right),$$

lowering sovereign default risk. Thus, with risk-averse banks and endogenous inflation, reserve rates open a channel through which monetary policy affects sovereign default risk via the real value of nominal debt. This channel is absent in the risk-neutral specification where inflation is indeterminate.

Comparison with a Taylor rule. A Taylor rule also pins down inflation rates (through a feedback rule on i_t^R) but leaves P_0 as a free normalization. Risk-averse banks and a Taylor rule therefore achieve the same degree of nominal determinacy, differing only in mechanism: goods-market clearing versus a monetary-policy feedback rule. Either alone suffices to pin down the inflation path.

However, the two mechanisms generate different comparative statics for the effect of reserve rates on inflation. Without a Taylor rule (exogenous i_t^R), the analysis above implies that a higher reserve rate raises inflation, because real rates are constrained by real fundamentals and cannot fully absorb the increase in i_t^R. Under an active Taylor rule ($\phi_\pi > 1$), the standard New Keynesian logic applies: the central bank raises i_t^R more than one-for-one in response to inflation, which increases the real rate and stabilizes inflation. In the first case, higher reserve rates reduce the real value of nominal debt and lower default risk. In the second case, higher reserve rates tighten financial conditions and may worsen borrowing terms.

A Limitation of the Current Setup and Two Minimal Fixes

A limitation of the current setup is that reserve issuance relaxes the government's period-0 financing need too easily. Combining the fiscal authority's period-0 budget constraint with the central bank's period-0 budget constraint gives

$$P_0 C_0^F = P_0 X + Q_0^B(B) B + T_0^{CB}, \qquad T_0^{CB} = M_0 - Q_0^B(B) B_0^{CB},$$

so that

$$P_0 C_0^F = P_0 X + Q_0^B(B)\big(B - B_0^{CB}\big) + M_0.$$

Holding central-bank bond purchases fixed, an increase in M_0 raises the fiscal authority's initial resources one-for-one and can substitute for risky sovereign issuance.

The problem is that the model makes reserve finance cheaper than sovereign debt finance for a given amount of period-0 resources. To raise one extra unit of resources with reserves, the consolidated public sector issues one extra unit of reserves today and owes the gross reserve return $(1 + i_0^R)$ next period. By contrast, to raise one extra unit of resources through sovereign bonds, the fiscal authority must issue face value $1/Q_0^B(B)$, so the gross financing cost is $1/Q_0^B(B)$. The bond-reserve funding wedge is especially transparent at time 1:

$$\frac{1}{Q_1^B} = \frac{1 + i_1^R + \tau_1}{1 - \bar{D}_1(B)} > 1 + i_1^R, \qquad Q_0^B(B) = \frac{E_0[Q_1^B \mid B]}{1 + i_0^R + \tau_0}.$$

Thus sovereign financing carries a higher yield than reserves because bond prices are discounted by default risk and, with risk-averse banks, by the additional premium τ_t. Reserve issuance therefore gives the consolidated public sector a cheaper funding instrument than risky sovereign debt.

The result is that reserve issuance looks too attractive: it finances period-0 spending at a lower effective cost than risky sovereign debt, yet the associated reserve liabilities do not tighten the government's future repayment-default tradeoff. Since reserves are voluntarily held, banks face no explicit reserve-holding friction, and the central bank can run negative net worth, sufficiently large reserve issuance can in principle eliminate the need for risky sovereign borrowing.

How the Literature Disciplines Reserve Issuance

Reserve demand. The IMF paper by Chen et al. [2023] is mainly an operational paper, not a sovereign-default or DSGE paper, but it provides a useful motivation for imposing a downward-sloping reserve demand curve. Its basic mechanism is a buffer-stock or liquidity-insurance story. Banks hold reserves because reserves are the most liquid asset in the system and help them absorb unexpected payment outflows, satisfy reserve or regulatory needs, and avoid costly short-term funding when liquidity is tight. As aggregate reserves rise, the marginal value of one more unit of reserves falls because banks are already better insured against settlement shocks. In a corridor system this implies that the short-term rate moves down toward the deposit facility rate as reserves become abundant, while when reserves are scarce banks are closer to borrowing at the lending facility and the marginal value of reserves is high.

> The demand for reserves describes the rate at which banks are ready to borrow and lend reserve as a function of aggregated reserves in the system. Banks seek to keep a certain buffer of reserves to absorb unexpected payments out of their account at the central bank. Their demand for reserves depends on the risk and predictability of payment shocks, interbank market functioning, the cost of liquidity-management errors, and reserve or regulatory requirements.

> With standing lending and deposit facilities, short-term rates are expected to decline as reserves increase, converging to the deposit facility rate.

The paper also stresses that more reserves are not always better. Excess reserves are low-yielding assets on bank balance sheets and can distort money-market intermediation, so there is a cost to holding too much liquidity. That is why the reserve-demand curve is not flat everywhere: reserves provide insurance, but with diminishing marginal benefit. For the present draft, the key implication is that a reserve-demand curve is naturally motivated by liquidity services, that it slopes down because the marginal convenience yield of reserves falls as reserve buffers rise, and that in practice it is often represented by a sigmoid or logistic schedule because rates are bounded by lending and deposit facilities.

Central-bank solvency. The motivation for a central-bank solvency or fiscal-support constraint is different. The relevant point in Del Negro and Sims [2015], Hall and Reis [2015], and Reis [2017] is that a central bank can always make a payment today by issuing more reserves, so solvency is not about running out of cash. The economically relevant question is whether the central bank can keep following its policy rule without reserve liabilities exploding or without being forced to generate extra inflation.

Del Negro and Sims [2015] give the clearest statement of the distinction between fiscal backing and fiscal support. Their argument is that even if fiscal policy backs the price level in the fiscal-theory sense, the central bank may still need Treasury recapitalization to keep hitting its inflation target if its balance sheet is badly impaired.

> Fiscal backing requires that explosive inflationary or deflationary behavior of the price level be ruled out by fiscal policy. But even when fiscal policy guarantees a unique price level and inflation is driven by monetary policy, it is possible that the central bank, if its balance sheet is sufficiently impaired, may need recapitalization in order to maintain its commitment to a policy rule or an inflation target. Such recapitalization is a within-government transaction, unrelated to whether fiscal backing is present.

Their extra step is important for the present note. If the central bank has a large long-duration balance sheet, then higher expected inflation or higher interest rates reduce the value of its assets.

If the Treasury will not support it, the central bank may need more seigniorage to cover the hole, and that can validate the inflation expectations. In that sense, solvency becomes a source of indeterminacy rather than a mere accounting issue.

Hall and Reis [2015] frame the issue more institutionally. They define central-bank financial stability by whether reserves stay stationary rather than drift upward forever. The key object in their analysis is the charter or dividend rule.

> A central bank can always meet its obligation to deliver the mandated dividend to the government because it has the unlimited power to borrow from commercial banks by issuing reserves. But a central bank cannot continue as a functioning financial institution independent of the government if it appears to be on a path of issuing an exploding volume of reserves. In that case, the central bank would be engaging in a Ponzi scheme.

So for Hall and Reis, the relevant solvency constraint is that if losses must always be financed by more interest-bearing reserves, and the institutional rules do not allow negative dividends, recapitalization, or make-up reductions in future remittances, then reserves can drift explosively. That is the relevant notion of insolvency.

Reis [2017] use the same logic as a limitation on quantitative easing. In his baseline, if the Treasury fully backs the central bank, losses are irrelevant for the consolidated government, so quantitative easing can be very large. But in a fiscal crisis that assumption is no longer innocuous.

> If the central bank can potentially receive payments from the fiscal authority when net income is negative, then it can implement quantitative easing while always staying solvent. In a fiscal crisis, though, the fiscal authority may not be willing to provide this fiscal backing to the central bank, which puts a solvency constraint on the central bank's actions, in the form of a lower bound on the possibly negative dividends to the fiscal authority.

So Reis's point is that without fiscal backing, negative remittances are no longer just an accounting mirror image. They become a genuine policy constraint that limits balance-sheet expansion.

Two disciplines. Taken together, these two disciplines operate on different margins. A reserve-demand curve limits reserve issuance from the private side, because banks stop wanting extra reserves unless they receive enough liquidity benefit or enough remuneration. A central-bank solvency constraint limits reserve issuance from the public side, because the Treasury cannot treat arbitrarily negative future remittances as harmless if the central bank cannot roll losses forever without reserve explosion or extra inflation. In the current draft, there is effectively neither discipline: banks can hold reserves freely with no explicit diminishing liquidity value, and the central bank can run negative net worth and make negative transfers with no fiscal-support limit. That is why reserve issuance currently looks too powerful.

A Downward-Sloping Reserve Demand Curve

A first minimal way to discipline this margin is to introduce a downward-sloping reserve demand curve directly in reduced form. A convenient way to do so is to let banks derive utility not only from consumption but also from real reserve holdings:

$$u(C_0^A) + \beta u(C_1^A) + \beta^2 \mathbb{E}_0[u(C_2^A)] + \chi_0 \ell\left(\frac{m_0}{P_0}\right) + \beta \chi_1 \ell\left(\frac{m_1}{P_1}\right),$$

where $\ell'(\cdot) > 0$ and $\ell''(\cdot) < 0$. Then the reserve Euler equation becomes

$$\frac{u'(C_t^A)}{P_t} = \frac{\chi_t \ell'(m_t/P_t)}{P_t} + \beta(1 + i_t^R)\, \mathbb{E}_t\left[\frac{u'(C_{t+1}^A)}{P_{t+1}}\right].$$

The extra term is the reduced-form counterpart of the shadow value λ_t in the payment-services microfoundation above. Because it is decreasing in reserve holdings, reserve quantities are no longer free: for a given reserve rate, there is a finite demand for reserves. Equivalently, one can summarize the same force through an inverse reserve-demand curve

$$i_t^R = \mathcal{I}_t\left(\frac{m_t}{P_t}, \text{state}\right), \qquad \mathcal{I}_t' > 0,$$

and let reserve quantities be pinned down endogenously by bank demand and reserve-market clearing rather than chosen independently by policy.

A simple payment-services microfoundation can rationalize this reduced form. Let z_t^N denote payment services used by each non-bank in period t, and suppose non-banks value these services because they facilitate transactions. Aggregate payment services are $Z_t = (1-\mu)z_t^N$. At the per-bank level, let

$$z_t \equiv \frac{Z_t}{\mu} = \frac{1-\mu}{\mu} z_t^N$$

denote payment services supplied by one bank. Suppose banks must hold reserves to settle these payments:

$$m_t \geq \theta P_t z_t.$$

Let s_t denote the real fee per unit of payment services paid by non-banks to banks. The fee applies only in periods $t = 0, 1$, because these are the periods in which reserves are held and used for settlement. Suppose $\mathcal{U}_C > 0$, $\mathcal{U}_z > 0$, and the marginal rate of substitution $\mathcal{U}_z/\mathcal{U}_C$ is decreasing in z_t^N. The non-bank problem can then be written intertemporally, in per-non-bank terms, as

$$\max_{\{C_t^N, B_t^N, z_t^N\}} \mathcal{U}(C_0^N, z_0^N) + \beta \mathcal{U}(C_1^N, z_1^N) + \beta^2 \mathbb{E}_0[u(C_2^N)]$$

subject to

$$P_0 C_0^N + Q_0^B(B) B_0^N + P_0 s_0 z_0^N = P_0 W_0^N,$$

$$P_1 C_1^N + Q_1^B B_1^N + P_1 s_1 z_1^N = P_1 W_1^N + Q_1^B B_0^N,$$

$$P_2 C_2^N = \mathbf{1}_2^R(B) B_1^N,$$

There is no period-2 service fee in this finite-horizon version because there is no period-2 reserve choice m_2 or reserve supply M_2.

Let ν_t^N denote the current-value non-bank marginal utility of one nominal unit in period t. The first-order conditions for C_t^N and z_t^N imply

$$\mathcal{U}_C(C_t^N, z_t^N) = \nu_t^N P_t, \qquad \nu_t^N P_t s_t = \mathcal{U}_z(C_t^N, z_t^N), \qquad t = 0, 1.$$

Combining these conditions gives the inverse demand for payment services,

$$s_t = \frac{\mathcal{U}_z(C_t^N, z_t^N)}{\mathcal{U}_C(C_t^N, z_t^N)}, \qquad t = 0, 1.$$

Thus the service fee is the non-bank marginal rate of substitution between payment services and consumption. If this marginal rate of substitution is decreasing in z_t^N, payment-service demand is downward sloping.

Now embed fee income directly in the bank's intertemporal problem. For $t = 0, 1$, the bank receives nominal payment-service revenue $P_t s_t z_t$ in addition to its endowment income. The bank solves

$$\max_{\{C_t^A, B_t^A, m_t, z_t\}} u(C_0^A) + \beta u(C_1^A) + \beta^2 \mathbb{E}_0[u(C_2^A)]$$

subject to

$$P_0 C_0^A + Q_0^B(B) B_0^A + m_0 = P_0 W_0^A + P_0 s_0 z_0,$$

$$P_1 C_1^A + Q_1^B B_1^A + m_1 = P_1 W_1^A + (1 + i_0^R) m_0 + Q_1^B B_0^A + P_1 s_1 z_1,$$

$$P_2 C_2^A = (1 + i_1^R) m_1 + \mathbf{1}_2^R(B) B_1^A,$$

and the settlement constraints

$$m_t \geq \theta P_t z_t, \qquad t = 0, 1.$$

Let λ_t denote the current-value multiplier on the period-t settlement constraint, measured in marginal utility units per nominal reserve. The first-order condition with respect to reserves is

$$\frac{u'(C_t^A)}{P_t} = \lambda_t + \beta(1 + i_t^R)\, \mathbb{E}_t \left[\frac{u'(C_{t+1}^A)}{P_{t+1}} \right], \qquad t = 0, 1.$$

The term λ_t is the shadow value of reserves coming from their role in settlement. The first-order condition with respect to payment services is

$$\frac{u'(C_t^A)}{P_t} P_t s_t = \lambda_t \theta P_t,$$

or

$$\lambda_t = \frac{u'(C_t^A)}{P_t} \frac{s_t}{\theta}.$$

To map this microfoundation into the reduced-form notation, let

$$x_t \equiv \frac{m_t}{P_t}$$

denote per-bank real reserve holdings. If the reserve requirement binds, then

$$x_t = \theta z_t = \theta \frac{1-\mu}{\mu} z_t^N, \qquad z_t^N = \frac{\mu}{\theta(1-\mu)} x_t.$$

Substituting into the fee equation gives the inverse demand for payment services as a function of real reserves:

$$s_t(x_t) = \frac{\mathcal{U}_z\Big(C_t^N, \frac{\mu}{\theta(1-\mu)} x_t\Big)}{\mathcal{U}_C\Big(C_t^N, \frac{\mu}{\theta(1-\mu)} x_t\Big)}.$$

Combining the inverse demand for payment services with the bank's payment-service first-order condition gives the induced inverse demand for reserves directly in terms of the shadow value λ_t:

$$\lambda_t(x_t) = \frac{u'(C_t^A)}{P_t} \frac{s_t(x_t)}{\theta} = \frac{u'(C_t^A)}{P_t} \frac{1}{\theta} \frac{\mathcal{U}_z\Big(C_t^N, \frac{\mu}{\theta(1-\mu)} x_t\Big)}{\mathcal{U}_C\Big(C_t^N, \frac{\mu}{\theta(1-\mu)} x_t\Big)}.$$

Holding the bank marginal value of nominal wealth $u'(C_t^A)/P_t$ fixed, differentiating yields

$$\lambda_t'(x_t) = \frac{u'(C_t^A)}{P_t} \frac{\mu}{\theta^2(1-\mu)} \frac{\partial}{\partial z} \left[\frac{\mathcal{U}_z(C_t^N, z)}{\mathcal{U}_C(C_t^N, z)}\right]_{z=\frac{\mu}{\theta(1-\mu)} x_t}.$$

This derivative is negative whenever the non-bank marginal rate of substitution $\mathcal{U}_z/\mathcal{U}_C$ is decreasing in payment services. Thus the shadow value of reserves declines with reserve holdings along the non-bank inverse demand curve. The reduced-form specification is obtained by defining

$$\frac{\chi_t \ell'(m_t/P_t)}{P_t} = \lambda_t\left(\frac{m_t}{P_t}\right),$$

so the convenience-yield term in the reduced-form reserve Euler equation is exactly the nominal counterpart of the shadow value generated by the payment-services microfoundation. The concavity condition $\ell''(\cdot) < 0$ is therefore the reduced-form representation of the declining marginal rate of substitution between payment services and consumption.

A Central-Bank Solvency Constraint

A second minimal way to discipline reserve issuance is to impose a central-bank solvency or fiscal-support constraint. A reduced-form analogue in the present finite-horizon environment is to impose a lower bound on period-2 remittances,

$$T_2^{CB,R} \geq -\bar{\Xi}, \qquad T_2^{CB,D} \geq -\bar{\Xi},$$

where $\bar{\Xi}$ summarizes the present value of future seigniorage, central-bank capital, or maximum fiscal support. Using the remittance equations above, this implies

$$B^{CB} - (1 + i_1^R)M_1 \geq -\bar{\Xi}, \qquad -(1 + i_1^R)M_1 \geq -\bar{\Xi}.$$

The default-state condition is tighter, so

$$(1 + i_1^R)M_1 \leq \bar{\Xi}.$$

Under this solvency bound, reserve issuance can still help finance the government, but only up to the amount that can be credibly serviced without unbounded future losses or explicit fiscal recapitalization.

Yang to Pedro: This way of imposing a solvency constraint seems to be too mechanical. It is no better to setting an upper bound for M_0.

Some Additional Observations

In the previous environments, we abstract from several institutional features to maintain tractability. In particular, the environments have the following implicit assumptions:

- Banks are not required to hold reserves. They choose reserves voluntarily as part of their portfolio, no binding constraints.

- Perfect interbank markets. As a result, banks can freely reallocate reserves and bonds. Thus, there are no: (i) interbank frictions, (ii) collateral constraints; nor (iii) funding constraints.
- No frictions in secondary markets fro public debt. Bonds can be traded freely between periods. Thus, there a no bid–ask spreads.
- No central bank capital constraint. Thus, the central bank can run negative net worth if necessary. Default-state transfers can be negative. Thus the CB balance sheet does not restrict policy.

These are not innocuous. In addition to the adding risk aversion in the banking sector, some the previous bullet points could be aspects that one could also consider analyzing.

References

Zhuohui Chen, Nikolaos Kourentzes, and Romain M. Veyrune. Modeling the reserve demand to facilitate central bank operations. IMF Working Paper 23/179, International Monetary Fund, 2023.

Marco Del Negro and Christopher A. Sims. When does a central bank's balance sheet require fiscal support? Staff Report 701, Federal Reserve Bank of New York, 2015.

Robert E. Hall and Ricardo Reis. Maintaining central-bank financial stability under new-style central banking. NBER Working Paper 21173, National Bureau of Economic Research, 2015.

Ricardo Reis. Qe in the future: The central bank's balance sheet in a fiscal crisis. IMF Economic Review, 65(1):71–112, 2017. doi: 10.1057/s41308-017-0028-2.